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SWE

Agile Model

* It was primarily designed to help a project to adapt to change requests quickly.
* The main aim of agile model is to facilitate quick project completion.
* To accomplish this task agility is required, which is further achieved by fitting the process to a project removing the activities that may not be essential for a specific project.
* Thus, anything that is wastage of time & effort is avoided.
* Agile model refers to a group of development processes. These processes share some basic characteristics but do have certain subtle differences among themselves.
* Some models of agile are -
  1. Crystal
  2. Atern
  3. Feature driven development
  4. Scrum
  5. Extreme programming (XP)
  6. Team development
  7. Unified Process

These are main ones.

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* The requirements are decomposed into many small parts that can be incrementally developed. Thus, adopting iterative philosophy.

* Each iteration is intended to be small & easily manageable that can be completed in a couple of weeks only.

* Phases of Agile-

(1) Requirement gathering
(2) Requirement Analysis
(3) Design
(4) Coding
(5) Unit Testing
(6) Acceptance Testing

* The time to execute an iteration is known as time box. Time box refers to the max^m amount of time needed to deliver an iteration to customers.

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* The central principle of an agile model is delivery of an increment to customer after each time box.

→ Principles Of Agile Model

(1) To establish close contact with the customer and to have the clear understanding of requirements, every team must have a customer representative.

(2) After each iteration, customer representative is responsible to review the progress of the project to revaluate customer requirement (to update goal).

(3) This model relies on working S/W deployment rather than comprehensive documentation.

(3) Requirement change requests from the customer are encouraged & efficiently ~~onto~~ incorporated.

(4) It incorporates use of efficient team members & effective communication among them using face-to-face communication.

formal ⇒ defined by mathematical rules

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(5) It uses the concept of pair programming where one programmer codes while other tests while it is being written.

→ Formal Model

↳ specifically used for scientific applications & projects.

Transformation model, specification driven model or formal method

⇒ Clear specifications are given since mathematical approach is employed.

⇒ Since clear specifications are available, verification becomes easier.

⇒ Since verification is easy, result is a defect-free product.

→ Concurrent Process Model

↳ processing in parallel

Client - Server { know how parameters }

Parallel system
↓
Centralized system
↓
Tightly coupled system

Distributed system
↓
Decentralized system
↓
loosely coupled system.

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Projects → embedded mode ⇒ complex apps / system S/W
↙ organic mode ⇐ small apps
↓ semi-detached mode ⇒ medium utility apps / software

* Summary ⇒ linear & sequential

(1) Waterfall Model
(2) Component Assembly / Object Oriented Model
(3) Formal Model
(4) Evolutionary Model ⇒ iterative philosophy

subcategories →
- Spiral
- WINWIN spiral
- concurrent process model
- Prototype
  - Evolutionary prototype ⇒ only 1 prototype
  - Throwaway prototype ⇒ n prototypes
  - RAD (Rapid Application Development) ⇒ multiple versions

for organic projects.

(5) Incremental Model

• Process Model

Projects → embedded mode
organic mode ← semi-detached mode
→ complex

formal ⇒ defined by mathematical rules

Summary

Waterfa
compone
Form
Evolution
→ Spir
→ WIN
→ Conc
mo
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Process

→ Concurrent Process Model:

Client Server

Parallel System
/
Centralized
/
Tightly Coupled

Distributed System
/
Decentralized
/
loosely Coupled System

→ Process Model

It describes the development & modification of steps and guidelines to achieve quality, continuous improvement, control process activities & to estimate the effort for a product

Microsoft → Streamlined Synchronizer

IBM → Rationalized, unified process Model

Infosys → Waterfall / Spiral

SUMMARY:

1. Waterfall
2. Component Based Model
3. formal Model / Specification Model

4. Evolutionary:
It utilizes the concept of
↓
1. Spiral
2. Win-Win Spiral
3. Concurrent process Model
4. Prototype → Evolutionary
→ Throwaway
→ RAD

Utility Software → Disk fragmen

Projects: Organic : Small
Semi-Detached
Embedded : Large

ral ⇒ defined by mathematical rules

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Utility Software → Disk fragmented

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4. Evolutionary:
It utilizes the concept of Iteration. (Iterative)
↓
1. Spiral.
2. Win-Win Spiral
3. Concurrent process Model
4. Prototype → Evolutionary prototype (1 prototype)
→ Throw away prototype. (n prototypes)
→ RAD (Rapid Application development)
↳ Software early phase development.

co

- Projects: Organic: Small projects, Application Software.
Semi-Detached. Medium, Utility Software.
Embedded: Large, System Software
↓
Compilers, Assemblers.